Digital Audio-Video Signal Processing

Prof. Marco Tagliasacchi

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1 Question - 7 pts

- Illustrate the basic concepts of scalar quantization.
- Briefly describe the Lloyd-Max algorithm.
- Define the concept of optimal rate-distortion curve of a source.
- Does the Lloyd-Max quantizer attain the optimal rate-distortion curve? Why?

2 Question - 7 pts

- Draw the block diagram of the JPEG image coding algorithm, briefly describing each functional module
- Explain the main improvements achieved by the newer JPEG2000 standard
- Consider an $N \times N$ image. Express the computational complexity of the transform module of both JPEG (block size $B = 8$) and JPEG2000. As for JPEG2000, assume that the separable wavelet transform is implemented as a PRFB, where filtering is iterated L times on the low-pass subband only. Assume that the both the low-pass and the high-pass filters have K taps.

3 Question - 6 pts

Illustrate the psychoacoustic effects of frequency and temporal masking, explaining how they are exploited in modern perceptual audio coders.

4 Question - 12 pts

Let $z(n)$ be a Gaussian memoryless source $\sim \mathcal{N}(0, 2)$. Generate one realization of the following process

$$x(n) = \rho x(n-1) + z(n) \tag{1}$$

Implement a DPCM encoder and decoder. Plot the operational rate-distortion curve for $\rho = 0.8$ and $\rho = 0.99$.